CHAPTER 6

HARD DISK DRIVE INSTALLATION & SUPPORT

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Chapter Objectives

Upon completion of this chapter, the student will be able to install:

- IDE/ATA & SATA hard disk drives.
- External hard disk drives
- SCSI disks & devices
- Software (Windows 7 & Applications).

Sub-topics

6.1 Installation of internal HDDs
6.2 Installation of external HDDs
6.3 Installation of SCSI disks & devices
6.4 Installation of Software

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6.1 Installation of internal HDDs

Preparing for installation

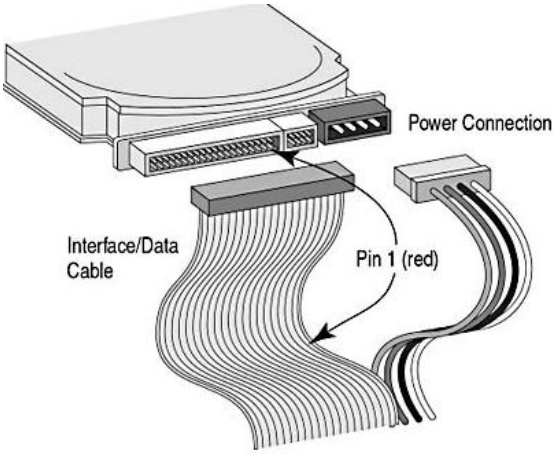
1. Convenient place e.g. clean (no dust), low-static (avoid carpets), flat surface.
2. Items required e.g. Phillips and flat-blade screwdrivers, grounding wrist-strap, notebook, and small container for (e.g. screws).
3. Computer manual i.e. documentation for that type of computer.

Precautions

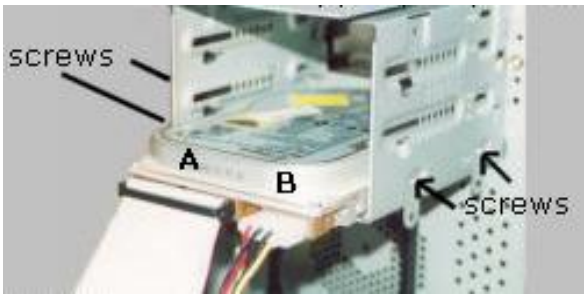
1. Turn off the power supply to PC, to avoid shock or even electrocution.
2. Discharge by grounding yourself, to avoid damaging microchips due to ESD.

How to install IDE/ATA hard disk drive

1. Remove the IDE data cable (ribbon cable) and the power cable.
2. Remove the 4 screws (2 on each side) or release plastic clips on the drive bay.
3. Gently take out the old hard disk drive out of the drive bay.
4. Set jumpers (master for single or slave for 2nd drive) on the new hard disk drive.
5. Gently slide the new hard disk drive into position.
6. Transfer the 4 screws from above and fix them on the new hard disk.
   (Plastic clips automatically lock it in position).
7. Insert the IDE data cable and power cable into the connectors.
8. Check in BIOS/CMOS setup whether 2nd HDD has been detected. Save the settings before you exit the BIOS/CMOS setup program. Alternatively, check in My Computer for drives listed. Remove the IDE cable & power cable



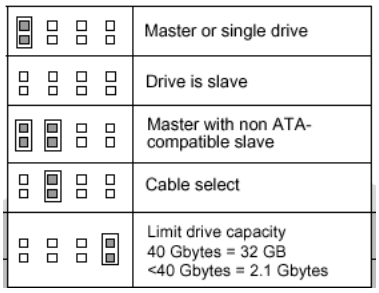
Remove the 4 screws



Hard disk in the bay

Hard disk fitted with 4 screws

Configure jumpers for master or slave
-Jumper positions vary depending on the manufacturer of the hard disk drive.
-Different hard drives have different jumper settings, so make sure to check labels printed on the hard disk for instructions about jumper positions.

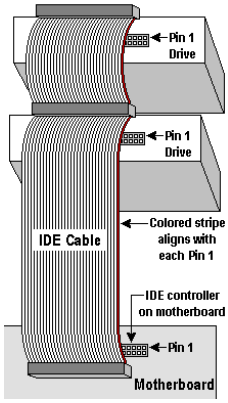


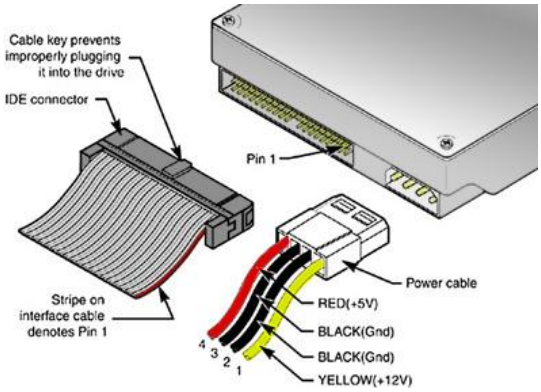
Note:

- Each of the 2 build-in IDE motherboard channels (connectors) can support up to 2 IDE/ATA drives (or optical drives).
- The terms **primary** and **secondary** refer to the 2 IDE channels, but also to the drives on each channel.
- We cannot have 2 master (or 2 slave) drives on the same IDE channel, because the PC will not boot.
- IDE primary drive on a channel is **master** and the other secondary drive is **slave**.
- IDE drives usually come configured as **master**; but you have to configure for **slave**.
- The boot drive may be set to **Cable select (CS)**; option for automatic configuration.

Connect IDE data cable & power cable

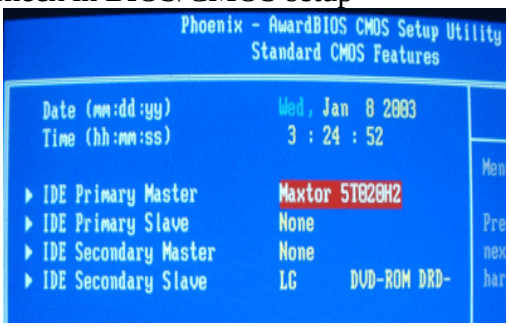
-The notch on the motherboard connector matches with alignment key on the connector.
-But if there is no notch, the red stripe (usually) is aligned with pin 1 on the IDE cable connector.

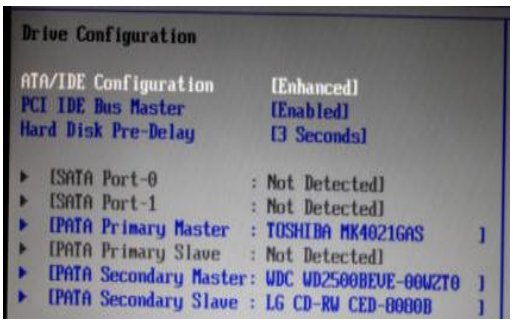




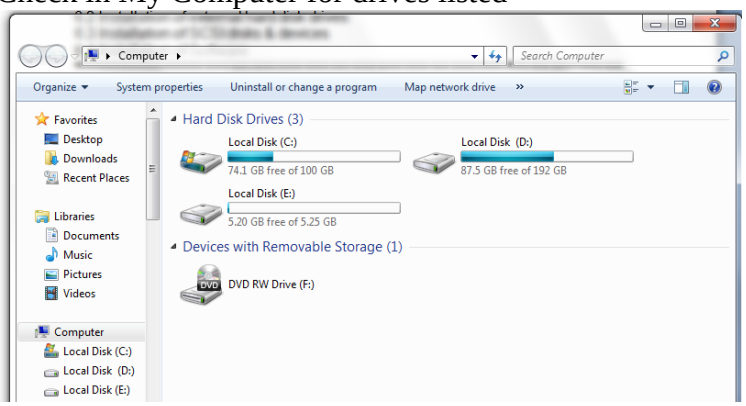
Why do you have to set master or slave?
-Each IDE drive has a controller on it; meaning 2 drives have 2 controllers to communicate with PC.
-Setting **master** selects one of the 2 controllers as active and to be used by all drives (master & slaves).
-Usually the primary master drive is the hard drive that the system boots from.

Check in BIOS/CMOS setup





-BIOS chips have IDE auto-detection mechanism i.e. a program which runs determines information about the drive e.g. manufacturer & capacity of the hard disk drive.
Check in My Computer for drives listed

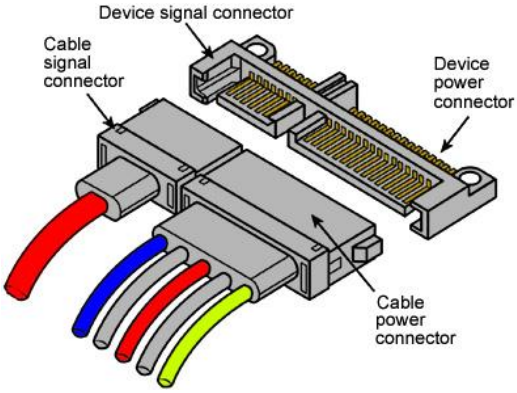


How to install SATA hard disk drive
-Similar to IDE/ATA; except setting jumpers for master/slave is **not** required with SATA drives.

-SATA hard drives also have jumpers, but their purpose is not the same as those of IDE and should be left with the default settings.
Check:

- The motherboard should have a SATA connector to connect SATA data cable. Newer motherboards can have both IDE/ATA connectors and SATA connectors.
- Each SATA motherboard connector supports only one SATA drive, compared to each IDE motherboard connector which supports 2 IDE drives/optical drives.
- The power supply unit should have a SATA power connector to connect SATA cable. If none, use special Serial ATA power cable to connect to power supply unit.
- SATA drives may have both 15-pin SATA power connector and 4-pin Molex IDE connector. Just use only them.

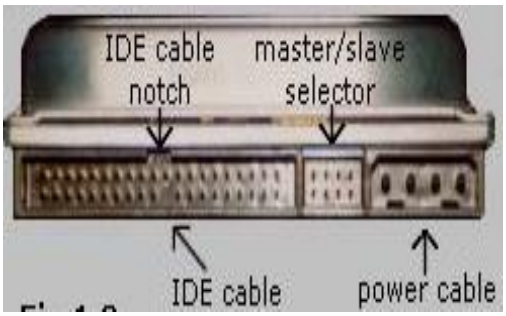
Connect SATA data cable & power cable

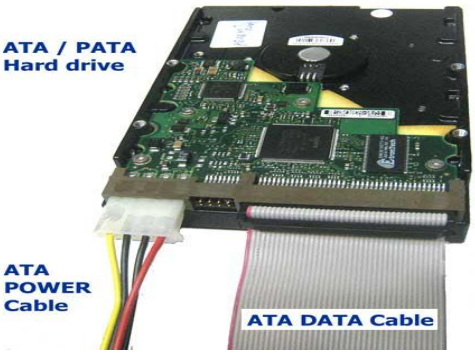


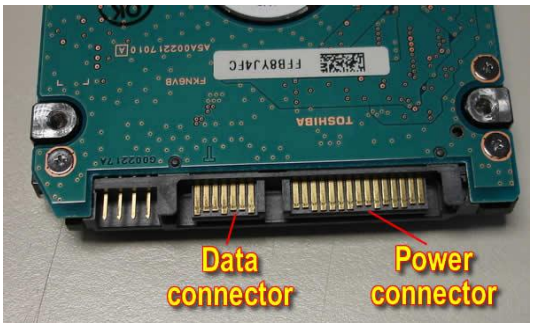
Question
Compare IDE/ATA and SATA hard disks in terms of:

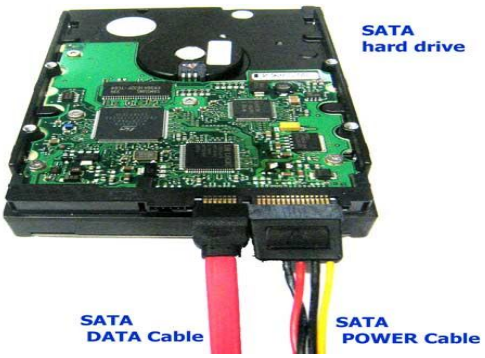
(a) Hard disk interfaces and their connectors
(b) Data cables and their connectors
(c) Power cables and their connectors
(d) Motherboard connectors

Hard disk interfaces and their connectors



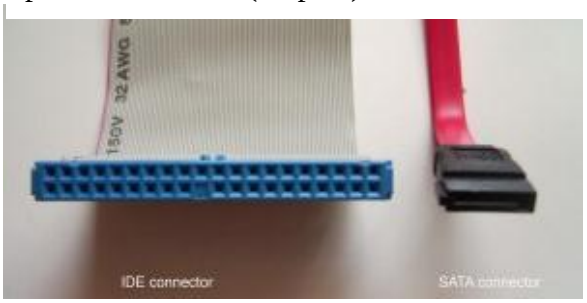






Top: IDE/ATA hard disk interfaces
IDE data cable connector (40 or 80 pins), jumper pins (6 to 10 pins) & power cable connector (4 pins).

Bottom: SATA hard disk interfaces
Jumper block (for factory use only), SATA data connector (7 conductors) & SATA power connector (15 pins).



Left: Wide 50mm IDE/ATA cable.

Right: Thinner 8mm SATA cable.

Both with alignment keys to match notches for installation.

Top: IDE/ATA cable has 3 connectors.

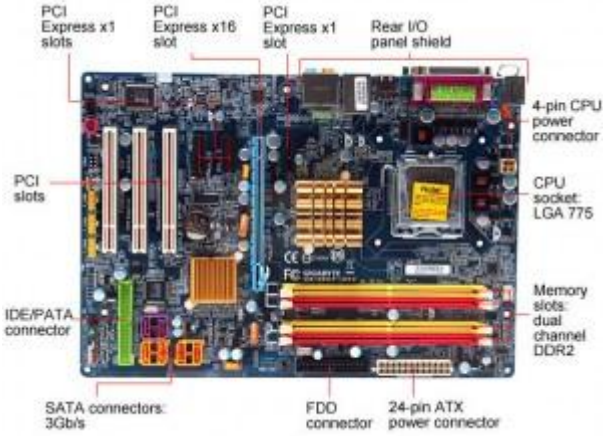
Bottom: SATA cable has 2 connectors.

Power cables and their connectors

Left:
Black 15-pin SATA power connector on power cable.

Right:
White standard 4-pin Molex IDE connector on power cable.

Motherboard connectors



Left Top:
Smaller /shorter SATA motherboard connectors. Each connects only one SATA drive i.e. SATA is **serial**.

Left Bottom:
Bigger/longer IDE/ATA motherboard connectors, each can connect 2 IDE drives i.e. IDE/ATA is **parallel**.

See also Chapter 5 for more diagrams.

Purchase special power cables for installation

1. Serial ATA power cable: If no SATA power supply with SATA hard disk
2. Serial ATA splitter power cable: If also IDE Molex connectors are needed.

L to R: Serial ATA power cable, SATA power cable input to 2 & 3 IDE/ATA connectors

Installation of special power cables

L to R: Serial ATA power cable to power supply, SATA power cable input 3 IDE/ATA disks

See also IDE to SATA (or SATA to IDE) 2 in 1 converter.

6.2 Installation of external hard disk drives
Types of external hard disk drives
- External hard disk drives differ depending on:
  1. Method of connection to PC i.e. connectors USB, FireWire, eSATA
  2. Storage of data into them i.e. types SSD, HHD, hybrid

- External hard disk drives also differ depending on whether they:
  1. Lie flat on sides or sit upright or both ways.
  2. Have stands which can be used or removed or others without stands.
  3. Are portable or desktop external hard disk drives (bigger & heavier).
  4. Provide some measure of extra security.
  5. Have very large capacities to automatically back up several PCs.

Storage of data
SSD (Solid State Drive) uses microchips for storage (has no moving parts).
HDD (Hard Disk Drive) uses a magnetic disk for storage and can get a little warm.
Hybrid is a mixture of both SSD and HDD characteristics.

SSD compared to HDD
- Works faster and runs cooler than the standard hard disk drive.
- It is more expensive and safer than HDD, because it has no moving parts.
- Usually has a limited amount of times that you can write it.
- Has reinforced exterior for added protection.

Method of connection to PC
Common connectors used are: USB, FireWire & eSATA
USB3.0 (newer version) is faster than USB 2.0 (older version).
eSATA is faster than USB 2.0, but usually requires external power source.

2 main methods of connections are:
1. External adapters i.e. uses external power of AC power adapter
2. Expansion cards i.e. uses internal power of PC

External adaptors
-This is the most common and safest, but expensive method used for both IDE/ATA and SATA hard disk drives e.g. it requires external power using AC power adapter.
-This can be done in a number of ways, which include:
  1. USB to IDE adapter cable
  2. USB to SATA adapter cable
  3. USB to IDE/SATA adapter cable
  4. eSATA to SATA adapter cable

Advantages of external HDDs
  1. Portability
  2. Security
  3. Backups

Installation of USB to IDE adapter cable

-USB connected to PC and the other end connected to IDE/ATA hard disk.
-External drive gets external power using by AC power adapter shown.

Installation of USB to SATA adapter cable

-Main USB converter port connected to PC when using USB cable for transfer.
-But if there is power shortage and the HDD fails to work properly, sub USB port is also connect to the PC.

Installation of USB to IDE/SATA adapter cable

-USB connected to PC and the other end connected to IDE/SATA hard disk.
-External drive gets external power using by AC power adapter shown.
See also eSATA to USB adapter combo cable.